



Counter-drone systems and data fusion

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2024

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EU Science Hub

<https://joint-research-centre.ec.europa.eu>

JRC139587

EUR 40146

PDF ISBN 978-92-68-22705-3 ISSN 1831-9424 doi:10.2760/6037951 KJ-01-24-196-EN-N

Luxembourg: Publications Office of the European Union, 2024

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How to cite this report: European Commission: Joint Research Centre, Grieco, G., Amendola, D. and Anderson, D., *Counter-drone systems and data fusion*, Publications Office of the European Union, Luxembourg, 2024, <https://data.europa.eu/doi/10.2760/6037951>, JRC139587.

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Abstract

The proliferation of Unmanned Aircraft Systems (UAS) leads to a wide range of legitimate, nuisance, and malicious applications. To address the growing concerns surrounding unauthorised drone usage, the Counter-Unmanned Aircraft System (C-UAS) technology has been developed. UAS are increasingly used in various sectors, necessitating effective C-UAS to mitigate potential threats. This report examines the complexities of C-UAS implementations, focusing on the detection, tracking, and identification (DTI) of unauthorised UAS in public spaces and critical infrastructures. Given the vast and diverse nature of sensors used in C-UAS, the study highlights the importance of multi-sensor data fusion for accurate DTI in real-time. It presents a taxonomy of sensors used in C-UAS, emphasising the need for a comprehensive, integrated approach to enhance system reliability and effectiveness. Moreover, it discusses the challenges of implementing a multi-sensor data fusion algorithm, with an overview of the current literature and the commercial solutions on the market. Finally, the report concludes with an update to the C-UAS proof of concept internally developed at the Joint Research Centre (JRC).

Foreword

This report describes the work performed by Giovanni Grieco during a five-month scientific traineeship¹ at European Commission's Joint Research Centre (JRC) in Geel, Belgium, between March and July 2024.

¹ More information about recruitment possibilities at JRC can be found at the following website:
<https://recruitment.jrc.ec.europa.eu>

Acknowledgements

The author expresses his sincere gratitude for the warm and proactive environment of JRC's Transport and Border Security research unit. JRC empowers a researcher's work with a clear mission: science for policy. Such vision enables the participation of independent research to the exemplar democratic system of the European Union, which has been felt from the first day to the last one of the traineeship. The importance of the work being pursued at JRC is critical for a solid and enduring democracy and for the advance of the European society. To this end, the author thanks his supervisors and his colleagues for such an opportunity and for a fulfilling experience.

– Giovanni Grieco, July 2024

1 Introduction

1.1 Societal context

Drones are sometimes referred to as unmanned aerial vehicles (UAVs) or remotely piloted aircraft systems (RPAS), but the term used in EU legislation ([European Commission, 2019a](#)) is unmanned aircraft systems (UAS). According to EU legislation, UAS refers to the whole system comprising pilot, operator, data-link, and unmanned aircraft (UA).

UAS are disrupting sectors of the aviation industry with affordable and easy access to airspace, influencing and impacting a wide range of applications ([Markarian et al., 2022](#)), e.g., urban air mobility, logistics, pharmaceuticals, last-mile delivery, emergency reconnaissance, agriculture, mapping, and surveying. UAS have unlocked an easy and affordable access to low-altitude airspace, further improving communications in overcrowded areas and in smart cities, where high-altitude non-terrestrial networks (NTNs), i.e. high-altitude platforms (HAPs) and satellites, are not as effective as drones and terrestrial networks (TNs). To this end, UAS are also becoming key actors for the realisation of future generation networks, sixth generation (6G) goals, as they blend together NTN services with TNs, as illustrated in Figure 1.

However, drones can also be used for illicit and hazardous activities, such as smuggling, harassment, trespassing private properties, accessing public sensitive areas, cyber-attacks, and terrorism ([European Commission, 2024b](#)). Addressing non-cooperative drone threats is crucial for legitimate drone usage. Drones can be classified according to their non-compliance nature (criminal, unlawful, or amateur). The growing use of drones in warfare theatres, recent terrorist attack plans, and the use of malicious drones demonstrate these risks. Suspicious drones near critical infrastructure and public spaces indicate potential hostile intelligence gathering. Criminals employ drones for smuggling and facilitating illicit operations such as drug trafficking and prison contraband.

EU legislation ([European Commission, 2019a](#); [European Commission, 2019b](#)) has been implemented to ensure safe flights and identify responsibilities. Beyond visual line of sight (BVLoS) flights are emerging as a key step towards autonomous flights, that is, without human intervention. These flights represent a stepping stone for the adoption of UAS at scale, beyond the current use cases for both private and enterprise use, monitored by unmanned traffic management (UTM) systems. However, this technology can easily be exploited by malicious actors, causing serious disruption to public spaces, crowded places, and civil buildings ([Counter Terrorism Preparedness Network, 2024](#)).

To this end, counter-unmanned aircraft system (C-UAS) solutions must be deployed in sensitive areas, in order to protect society and critical infrastructure. C-UAS implementations carry out detection, tracking, and identification (DTI) of unauthorised UAs accessing the protected area, also known as the “scan area”. They are able to do so thanks to a combination of different technologies, ranging from electromagnetic spectrum monitoring to audio sensing. While civil C-UAS implementations typically focus on DTI, military ones normally extend this concept to provide counterattack features for UA neutralisation and forensics procedures for post-hoc analysis and intelligence ([Krueckemeier et al., 2021](#)). While C-UAS is a topic in its infancy, multiple problems and solutions can be translated from other domains, such as traditional aviation, satellite imaging ([Di & Yu, 2023](#)), biomedical, automotive, and internet of things (IoT) ([Wang et al., 2021](#)). C-UAS solutions deal with vast amount of data; they handle a large data volume across a variety of sensors, which can be generated at different velocities, provide a form of value toward the success of DTI, also influenced by their quality and accuracy, i.e., veracity ([Di & Yu, 2023](#)). Furthermore, data are (i) from multiple sources of different nature, (ii) high-dimensional, especially for hyper-spectral sensors, (iii) subject to multiple scales and different spatial

resolutions, and (iv) isomeric, i.e., drones may be represented under different data structures, such as vector and raster. Moreover, their relationship can be non-linear. These aspects introduce networking and synchronisation problems to guarantee sufficient bandwidth and timing for the elaboration of big data. Furthermore, for distributed and wireless sensors, C-UAS inherits the issues related to remote sensing, further discussed in [Di & Yu \(2023\)](#).

Figure 1. Overview of T/NTN architecture for 6G communications.



Source: JRC

1.2 EU policy context

- The European Union (EU) has adopted common rules in the field of civil aviation that include the use of UAS in European skies ([European Commission, 2018](#)).
- Further measures were established under the Commission Delegated Regulation (EU) 2019/945 on unmanned aircraft systems ([European Commission, 2019a](#)) and Commission Implementing Regulation (EU) 2019/947 on the rules and procedures for the operation of unmanned aircraft ([European Commission, 2019b](#)).
- The EU has taken significant steps to address the security threats posed by drones, including the Action Plan to support the protection of public spaces ([European Commission, 2017](#)), the EU Security Union Strategy ([European Commission, 2020a](#)) and the Counter-Terrorism Agenda ([European Commission, 2020b](#)).
- EU funds have been allocated to research projects on drones, including on C-UAS ([European Commission, 2024b](#)).
- In 2023, the Commission adopted a new Communication specifically addressing the unlawful and dangerous use of civil drones ([European Commission, 2023c](#)).

1.3 Literature review

While C-UAS research and commercialisation are rapidly expanding (COURAGEOUS, 2022; Kashi et al., 2024; Lykou et al., 2020; Melnichuk et al., 2020; Park et al., 2021; Rauschen et al., 2019; Wang et al., 2021), current state of the art has its foundations in Blackman & Popoli (1999), which has been of great use in the aviation industry to DTI multiple aircraft. Furthermore, such principles can be extended to UAS vulnerabilities and their potential intrusion strategies (Hengy et al., 2017) which are key for their DTI (Krishna & Murphy, 2017).

The European Commission's Joint Research Centre (JRC) is developing C-UAS knowledge and capability via implementation of its DRONE project (European Commission, 2022) and has published two handbooks on physical hardening (European Commission, 2023a) and protection of critical infrastructure and public spaces (European Commission, 2023b). In addition to setting up a Living Lab² in Geel, Belgium, the JRC has a framework contract for C-UAS services for research and protection of infrastructure of Commission premises in Geel and Brussels.

The COURAGEOUS project³, launched in 2021, focuses on developing a standardised test methodology to DTI malicious drones in order to protect the low-altitude airspace. The project aims to create a qualitative and quantitative comparison between different C-UAS by establishing a series of standard user-defined scenarios representing a wide range of use cases. By addressing the need for a standardised test methodology, the COURAGEOUS project significantly contributes to advancing the field and ensuring effective protection of low-altitude airspace from unauthorised drone usage. In (COURAGEOUS, 2022) the authors provide an overview of the trends for the design and the implementation of C-UAS solutions. Radiofrequency (RF), radio detection and ranging (RADAR), and visible-light sensors are preferred, whereas acoustic, light amplification by stimulated emission of radiation (LASER)-based, and infra-red (IR) ones are less common. While RF and RADAR are more likely to be used standalone, some solutions prefer the combination of RADAR, visible-light, and thermal sensors.

A comprehensive overview of cooperative scenarios, in which a C-UAS implementation is augmented with the use of UAS for patrolling, can be found in Castrillo et al. (2022). That publication also addresses neutralisation techniques in order to mitigate airspace breaches. A crucial aspect for this extended system is a robust wireless network that does not interfere with the counter system and is able to defend itself from external attacks. The challenges in these aspects can be found in several articles (Boccadoro et al., 2021; Krishna & Murphy, 2017; Yuvaraj & Velliangiri, 2023). More generally, in Laoudias et al. (2018), the authors provide a review on network localisation, tracking, and navigation, in particular on cellular localisation systems and solutions based on wireless local area networks.

Deep learning (DL) is also enhancing DTI in recent C-UAS implementations, as reported in Al-Iqubaydhi et al. (2024), Samaras et al. (2019), and Saqib et al. (2017). Among the several DL architectures, the multi-layer perceptron neural network, well known for its classical input, several hidden, and output

² Living labs are a modern way of creating user-centred environments that enable innovation, co-creation and start-up development. https://joint-research-centre.ec.europa.eu/living-labs-jrc_en

³ The COURAGEOUS project is funded under EU's Internal Security Fund Police (ISFP) under Grant Agreement 101034655, <https://courageous-isf.eu>

layers, is the most used for this application ([Gardner & Dorling, 1998](#)), especially for RADAR-based DTI ([Wang et al., 2019](#)).

It is critical for such DL systems to be as efficient and as accurate as possible for the DTI of unauthorised UAs. To this end, the development of algorithms able to do DTI in challenging conditions is a must to ensure the reliability of the system ([Di & Yu, 2023](#)). Currently, it is challenging to have reliable and fast DL models if the data is subject to imbalanced and noisy labels. Furthermore, the data must be partitioned, loaded, and the pipeline analysis task properly scheduled to balance the computing and networking resources, and thus keep the entire system under the real-time requirement for a prompt protection of the scan area. For an efficient use of the DTI models, proper feature selection has to be made to filter, embed, and wrap data before processing. Consequently, machine learning and/or DL models need to be adapted and tuned to the necessities of C-UAS solutions, especially in terms of fast data ingestion, their processing, and the ability to work with a continuous data stream. Data may be missing or noisy, as not all sensors are in their ideal conditions in a certain time frame, especially caused by natural phenomena, e.g., fog, mist, or direct sunlight. Furthermore, imbalanced data impact classifiers, for which proper techniques must be implemented to mitigate the issue. Finally, on-premises solutions can be part of a distributed and/or federated system of C-UAS implementations all-over the world, e.g., the ARTAS system in the case of airspace monitoring for traditional aviation ([Hoogendoorn & Neven, 1997](#)), in order to improve models through transfer learning and / or provide a broader early warning system to protected areas that are close together.

Given the implications of such systems in public and sensitive areas, regulations are being developed around the world, particularly in the European Union, with the UAS regulatory framework based on [European Commission \(2019a\)](#) and [European Commission \(2019b\)](#) and the United States with the Federal Aviation Administration (FAA) regulations ([Patel & Rizer, 2019](#)). Furthermore, preliminary information can be seen for the integration with the upcoming Air Mobility market ([Dulia & Shihab, 2024](#)). Given all these aspects, C-UAS design and sensor implementation are critical aspects for their DTI of UAS.

1.4 Scope of this report

This report highlights the recent advancements and the open challenges associated with creating a comprehensive solution for C-UAS. It also provides an updated taxonomy of sensors that can be employed in a C-UAS solution, outlining in section 2 their features and limitations, highlighting the need for multi- sensor data fusion framework for reliable and accurate DTI of UAS.

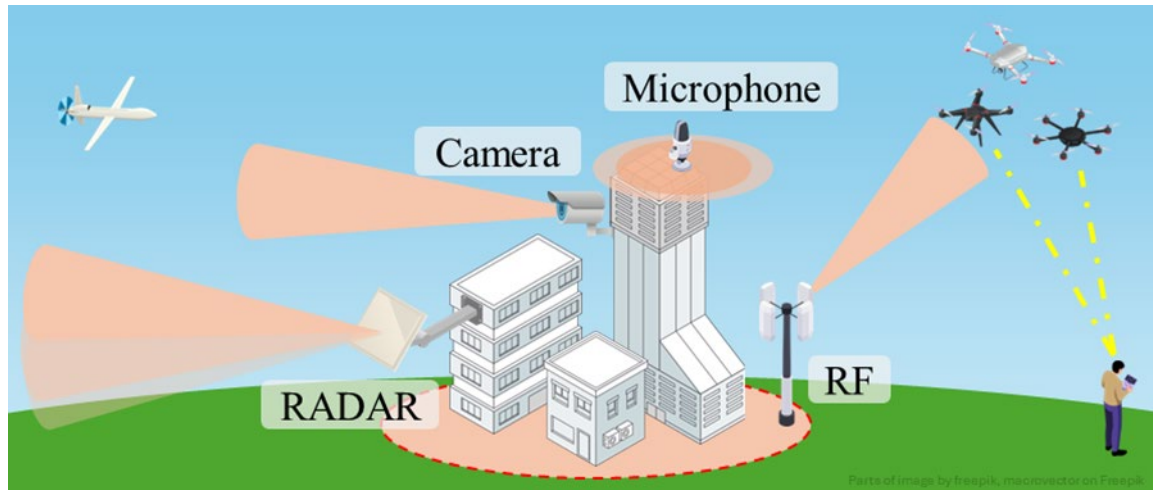
In section 3, a meta-study analysis of the scientific literature is presented concerning the state of the art and challenges regarding multi-sensor data fusion algorithms for C-UAS. The study has been limited to the DTI task, considering also the classification of UAS, as it fits the scope of C-UAS employed for civil purposes.

Section 4 of this report describes the complete code restructuring of the C-UAS Proof of Concept (PoC) developed within the project, and its adaptation to JRC's DevOps infrastructure.

2 Overview of C-UAS Sensors: Taxonomy and Challenges

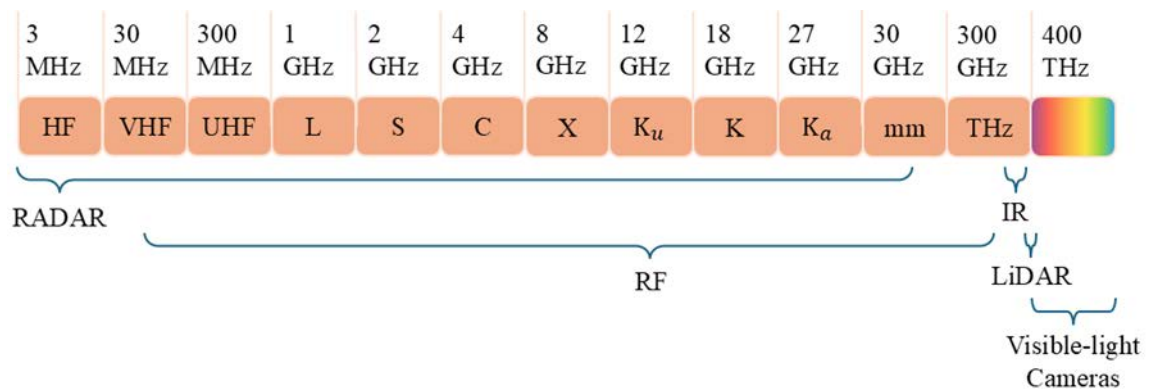
Sensors can be categorised in different ways (Di & Yu, 2023), as they (i) sense electromagnetic waves or mechanical ones; (ii) work on specific bands of the spectrum; (iii) analyse a signal being generated by the sensor itself or scan for signals propagating in the environment, and thus be classified as active or passive, respectively; (iv) operate in different working conditions; and (v) be installed on different platforms. A general overview of a C-UAS system is depicted in Figure 2, while a sensor categorisation employed to monitor the electromagnetic spectrum is illustrated in Figure 3.

Figure 2. Example of C-UAS sensors employed to defend a sensitive area, highlighted dashed red border.



Source : JRC

Figure 3. Overview of the types of sensors used to monitor the electromagnetic spectrum. Spectrum bands according labelled according to the relevant IEEE standard (IEEE, 2019).



Source: JRC

Each combination of these features designates a subset of sensors that brings a trade-off in terms of detection accuracy, cost of implementation, latency, and cost of computational resources, and thus it affects the quality of the data and influences the overall performance of the C-UAS solution.

Finally, sensors can be classified by their platform of reference (Di & Yu, 2023). In-situ sensors can be fixed, mounted on a tower or placed on the ground. They can also be mobile, as they can be mounted on cars or carried by humans. Sensors can also be airborne, as they can be mounted on

collaborative UAs, i.e., drones that extend and enhance the C-UAS DTI task. Given their variable altitude, their data acquisition may be subject to less clutter and interference.

2.1 RADAR

There are several types of RADARs, which offer different trade-offs in terms of performance, technological readiness, economic feasibility, and limitations. These sensors analyse a radio signal that has been initially transmitted by the sensor itself and then backscattered by the environment of the scan area and, possibly, by the targets of interest. In typical scenarios, the amplitude and phase shift of the returned signal describes the intensity and distance of the target from the sensor, respectively. Furthermore, the Doppler effect can be analysed, providing an estimation of the target velocity and its distance from the sensor. A specific study that has been made for the application of RADARs for C-UAS implementations can be found in [Brown \(2023\)](#) and [Wellig et al. \(2018\)](#), with current mmWave experimentations summarised in [Kong et al. \(2024\)](#) and micro-Doppler signatures discussed in [Tahmouh \(2015\)](#).

A key advantage of RADAR is that it tends to work better than the other types of sensors across diverse weather conditions, even though it is affected by the attenuation of rain, i.e., the heavier the rainfall and the higher the operating frequency, the higher the signal attenuation experienced (COURAGEOUS, 2022).

UAS can be of various size, with a wingspan that greatly ranges from 61 m to 1 mm ([Hassanalian & Abdelkefi, 2017](#)). To this end, the sensor choice would be firstly focused on their covered wavelength, which typically varies between 1 mm, i.e., 300 GHz, and 100 m, i.e., 300 MHz ([Di & Yu, 2023](#)). Current market trends (COURAGEOUS, 2022) indicate that a significant number of C-UAS implementations do not report the range capability of their RADAR, i.e., 45 %. The solutions focus more on DTI UAS in the 3~5 km and 7.5~10 km ranges.

Even though RADAR sensors are compelling to be included in a C-UAS implementation, considerations related to the spectrum regulation, the maximum power allowed in a certain area, interference, blind spots, resolution, and position are key aspects that can heavily influence their overall performance (COURAGEOUS, 2022).

Among several RADAR types and techniques to DTI UAS, automatic target recognition (ATR) ([Jiang et al., 2023](#)) emerges as a theoretical proposal to identify UAS characteristics from the echo signal and then further classify it. Even though ATR has achieved significant progress thanks to DL-based approaches, there are open challenges to overcome and real-world testing needed, among them the need for a dataset large enough for proper detection and recognition. Furthermore, with explainable AI (XAI), explainable models are in development for secure, trustworthy, and robust DTI.

Conversely, synthetic aperture RADARs (SARs) represent the most common RADARs. Their main advantage is the possibility of being installed on movable platforms, and thus it provides an enhanced aperture of the acquisition, hence its name. Furthermore, they provide high-resolution acquisitions independent of the time and weather condition. Among SARs, ATR is also being considered ([Li et al., 2023](#); [Li & Peng, 2023b](#); [Weng, 2023](#)). Another major challenge highlighted is the real-time requirement of C-UAS, which is often neglected in research and imposes strict limits on the complexity of the neural network, the ability to be compressed in smaller forms, and the ability to quickly elaborate data fragments acquired in a brief time interval. Furthermore, it is possible to obtain high resolution images through static synthesis of aperture and by imposing a circular movement to the sensor to obtain the best results ([Kolesnikov et al., 2023](#)).

2.2 RF analysis

An unmanned aircraft (UA), or unmanned aerial vehicles (UAV), is typically part of a much broader and rich UAS system, with the intent to offer remote control, telemetry, and application data stream to a remote pilot or a ground control station (Wang et al., 2021). To this end, UAs are subject to electromagnetic radiation for both the transmission and reception of signals from/to the remote pilot as well as Internet and localisation services, such as global positioning system (GPS). To this end, multiple software defined radios (SDRs) can be employed to detect the radio activity of UAs and determine its location through direction of arrival (DoA) or time of flight (ToF) measurements. Signal processing algorithms have been proposed for an accurate direction finder requiring just two pairs of antennas (Zhyla et al., 2023).

Traditional RF-based analysis techniques and Wi-Fi signal interception are discussed in Bisio et al. (2024), where it is highlighted that it is a challenge to eavesdrop Wi-Fi traffic in open field, especially if the scan area is larger than 3 km of radius, requiring expensive high-gain antennas or ad-hoc networks. Furthermore, there is the need to build a comprehensive dataset of communications recorded from different models of drones, in order to ease their detection and classification. The current state of the art focuses on laboratory tests with one single drone being analysed, thus severely limiting the research for a C-UAS solution able to DTI multiple UAS at long distances and to discriminate between legit drone communications, other types of communications, and fake traffic to confuse the C-UAS detectors. Finally, circumvention of Remote ID (Aouladhadj et al., 2023; ASD-STAN, 2021; Choi & Kang, 2023) represents another major challenge, in particular in relation to privacy safeguarding and regulation.

Passive RADAR methods for detection of drones are possible with SDRs and other receivers integrated in a heterogeneous bistatic RADAR setup. This is most useful in case of autonomous drones with no radio activity. These methods process reflections from non-cooperative sources of illumination, e.g., Wi-Fi, radio, DVB-T signals, in the environment to detect or localize moving objects. However, it should be noted that these methods do not specifically discriminate drones from other moving objects, which can result in false alarms. In this case, third-party radio activity such as nearby Wi-Fi stations, radios, and cellular network can be analysed, along with their interference, to understand the presence of autonomous UAs. In this regard, for example Deng et al. (2024a) demonstrated that their solution is able to DTI drones using channel state information (CSI) from Wi-Fi with interference to such signals caused by shifting, moving, and spinning propellers of a drone.

Many kinds of signals are transmitted over-the-air, one signal that is present all around the globe is GPS (Di & Yu, 2023), emitted from a constellation of satellites to provide global navigation satellite system (GNSS)-based location. Reflectometry techniques can help to analyse the surrounding environment, and thus to detect moving objects like UAs.

Simultaneous transmission and reception at the same frequency would ease RF monitoring deployments, as they require less hardware, e.g., SDRs, to transmit and then analyse the surrounding interference, working in full-duplex mode filtering out self-interference (Vihelä et al., 2024). For this case, tests were conducted in the military sector in the 225~400 MHz range up to 100 W of transmission power and in the civil cellular infrastructure for in-band 5G communications at 2.4 GHz (De Silva et al., 2024).

DL-augmented neural models provide superior DTI accuracy (Inani et al., 2023), as it has been demonstrated that it is possible to differentiate between different drone types, as well as their flight setup. This is due to the fact that it is possible to extrapolate frequency-related features that contain more information about the incoming UAs in respect to other types of sensors.

2.3 Electro-optical sensors

Active sensors in this category are typically dominated by the presence of light detection and ranging (LiDAR) devices, which employ pulsed laser beams for ranging (Di & Yu, 2023). Inversely, passive sensors of this category are further distinguished between common visible-light cameras, IR/thermal cameras, multi-spectral, and hyper-spectral, with the last two focusing on a few discrete bands and many narrow and contiguous bands, respectively (Di & Yu, 2023). Moreover, these cameras may be augmented in a pan-tilt-zoom (PTZ) setup, allowing three degrees of freedom in terms of movement and zooming (COURAGEOUS, 2022).

Electro-optical sensors may benefit from the recent advances in DL-aided computer vision systems. A comprehensive overview of models applied to these sensors can be found in Amjoud & Amrouch (2023). These proposals are severely limited in C-UAS solutions (COURAGEOUS, 2022) due to:

- their dependency to light, especially during nighttime;
- the lens choice, which imposes a certain focal length and the maximum zoom distance;
- the sensor resolution, which limits their detection range;
- adverse weather conditions;
- dirt on the sensor.

As for light sensitivity and adverse weather conditions, short-wave IR (SWIR) offers promising solutions (Castrillo et al., 2022). Moreover, it is challenging to accurately determine the UA coordinates, which may require at least two cameras in stereo configuration, a recent DL model for monocular ranging (Jin et al., 2023), or the coupling of a LASER for ranging.

Finally, the PTZ benefits are limited by their speed of movement and variable blind spot when zooming (COURAGEOUS, 2022). Among these difficulties, there is also the need to find the right trade-off in terms of timing and accuracy. Furthermore, the need for a large dataset may be compensated with few-shot learning, allowing the models to detect and classify new drone models via a few examples (Amjoud & Amrouch, 2023).

2.4 Mechanical wave analysis (acoustic sensors)

Sound navigation and ranging (SONAR) is not suitable for detecting and tracking targets using sound waves. Sound waves travel much more efficiently in water than in air, as water is denser than air and provides better acoustic coupling. Consequently, sound waves quickly lose energy in air, significantly limiting the effective range of SONAR.

However, microphone arrays have been successfully used for DTI given the typical rotor blade noise of UAs (Wang et al., 2021). An array of microphones is deployed inside and around the scan area to evaluate the time delay of arrival (TDoA) in real-time. Even though the system offers high accuracy detection and localisation in ideal conditions, it is severely limited by adverse weather conditions and environmental noise, which typically surpasses the rotor noise produced by UAs. To this end, acoustic-based DTI, both in its active and passive forms, are not typically favoured for C-UAS implementations (Wang et al., 2021) but should be used in specific conditions.

3 Multi-Sensor Data Fusion for C-UAS

3.1 Introduction

Given the limitations of individual sensor types, a C-UAS solution that relies solely on one type of sensor would be significantly constrained in its monitoring capabilities ([Seidaliyeva et al., 2024](#)). A more effective approach involves adopting a system architecture that benefits from multiple sensor types and implements a multi-sensor data fusion algorithm. This approach integrates the features of different sensors to enhance the overall accuracy of DTI, even when individual indicators are incomplete or uncertain ([van den Broek et al., 2019](#)).

Data fusion can be applied at different levels:

- at feature level, aiming to integrate different sensors data;
- at the decision level;
- at the rule level, which is the simplest approach for integrating a variety of sensors but sacrifices flexibility and adaptability ([Seidaliyeva et al., 2024](#); [Wang et al., 2024a](#)).

Multi-sensor data fusion presents an ongoing challenge, with the goal of providing a unified representation of these data within C-UAS implementations (ALADDIN project, 2020; [Samaras et al., 2019](#)). Major challenges in multi-sensor data fusion solutions include addressing sensor heterogeneity, managing misalignments between data acquisitions, and handling the high volume, variety, velocity, and veracity of multi-modal data, which often exhibit complex inter-modal correlations ([Wang et al., 2024b](#)).

Moreover, it is essential to identify a combination of sensors and classifiers that can produce consistent results or support DTI, even if some sensors fail to detect certain targets (e.g., autonomous UAS immune to passive RF monitoring). The goal is to avoid combinations that weaken the model's effectiveness for DTI of UAS. Currently, C-UAS solutions on the market combine RADAR with electro-optical cameras for independent tasks and integrate different types of cameras or RADAR with passive RF sensors to improve accuracy on the same tasks. Although qualitative analyses have been conducted by the research community to evaluate overall system reliability (COURAGEOUS, 2022), quantitative analyses that accurately and fully characterise an effective C-UAS solution remain scarce.

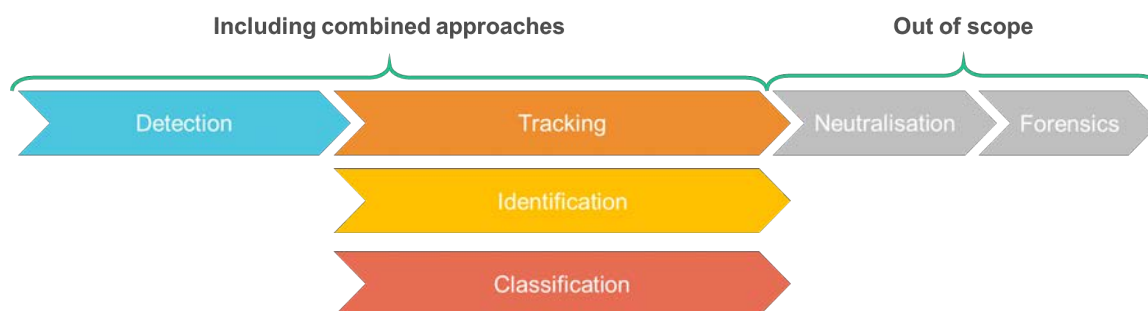
Despite the potential benefits, current C-UAS solutions on the market often lack transparency regarding the specific sensor fusion techniques employed. While some may briefly mention the use of machine learning or traditional statistical methods, the limited disclosure hampers a comprehensive understanding of their performance and potential areas for improvement. The combination of multiple sensors is a less common practice, and it is hindered by the complexities involved in merging and fusing multi-sensor data within such sophisticated systems (COURAGEOUS, 2022).

3.2 Meta-study analysis of the scientific literature

A meta-study analysis of the scientific literature has been conducted to evaluate the state of the art and current challenges regarding multi-sensor data fusion algorithms for C-UAS. The study has been limited to the DTI task, considering also the classification of UAS, as it fits the scope of C-UAS employed for civil purposes. Furthermore, the study was further limited to “static C-UAS” techniques,

as in C-UAS implementations that mainly used sensors fixed in place, instead of mobile, human-packable, and collaborative aerial ones (Kang et al., 2020).

Figure 4. Limitations imposed for the scientific literature analysis.

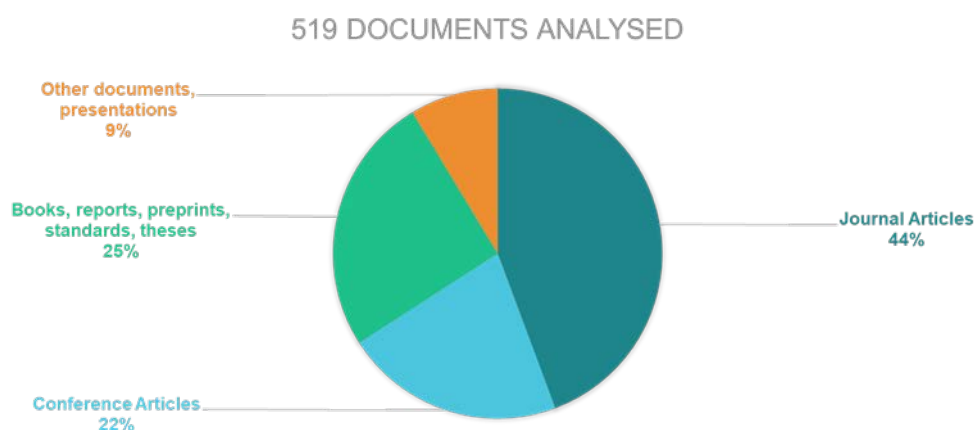


Source: JRC

The limitations of the analysis are illustrated in Figure 4, further including the fact that the tracking, identification, and classification tasks can be done after a UAS has been detected and may be carried out in parallel. In total, 519 documents have been retrieved through the European Commission's eLibrary and CORTEX tools, which allowed the access to works published in book, conferences, and journal articles, as well as EU-funded project deliverables and standards. As can be seen in Figure 5 and Figure 6, most of the documents analysed were journal articles, accounting for a total of 228 documents. Although foundational techniques to DTI aircraft have been researched since 1980, most of the literature focused on C-UAS was produced in the last four years, with 80 journal articles published in 2023 alone.

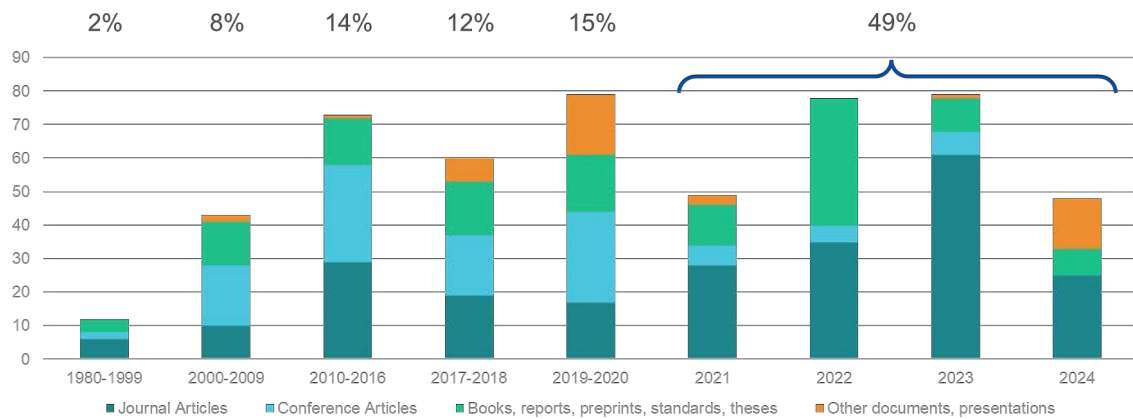
A fundamental takeaway from such a literature research is that the majority of C-UAS prefer the use of single type of sensor, even though multi-sensor systems are reportedly growing over time. An open challenge is finding ways for real-time multi-sensor data fusion, as it is currently one of the major limitations of the system scalability over the number of sensors and their types to be deployed. Furthermore, it is not always demonstrated convincingly that an increase in the number of sensors delivers an effective increase in accuracy and precision in DTI tasks.

Figure 5. Overview of the analysed documents per category.



Source: JRC

Figure 6. Temporal analysis of the analysed documents, per type.

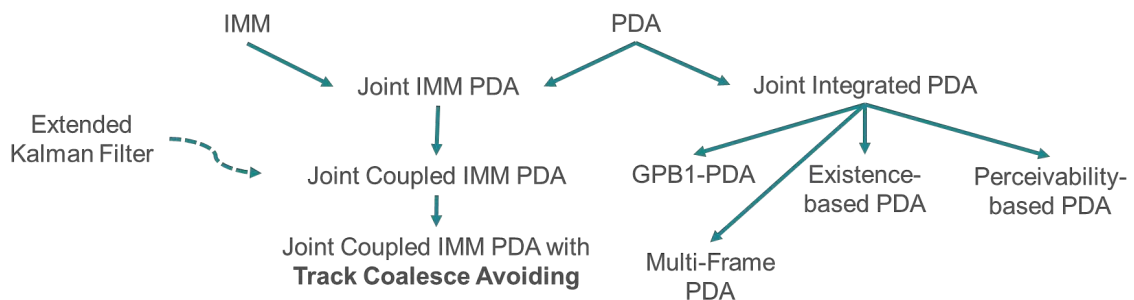


Source: JRC

Multi-sensor data fusion algorithms rely on key assumptions related on how architecturally close they are to their specific sensor features, i.e., early versus late fusion approaches (Samaras et al., 2019). This allows the multi-sensor data fusion to be treated as a composition of multiple models, promoting the use of DL-oriented models to analyse sensor raw data of a single type of sensor, e.g. YOLO for detection in electro-optical sensors (Deng et al., 2024b), together with high-level models that focus on tracking and decision-making based on pre-processed and filtered features.

Current proposals can be divided into three major categories: a classical approach, which relies on traditional Bayesian probability, a modern one, mainly focused on the multi-layer perceptron neural network, and a hybrid version that mixes both approaches. Classical multi-sensor data fusion is based on tracking techniques that can be found in the aviation industry, especially from the efforts done by the Netherlands Aerospace Centre (NLR) to create the ATLAS system, currently employed to protect the European airspace. These are algorithms based on interacting multiple model (IMM) and probabilistic data association (PDA), as illustrated in Figure 6, plus multi-hypothesis tracking (MHT) and clustering.

Figure 7. An overview of PDA-based approaches for C-UAS DTI tasks



Source: JRC

The main issue of these algorithms is their complexity, which hinder their possibility to be executed in real-time, and their risk on merging close (but distinct) tracks. As illustrated in Figure 6, IMM and PDA are typically merged in order to get the advantages of the two, enhanced with safety mechanisms to avoid unwanted track merge, even under partial and missing data.

Conversely, MHT is one of the pioneering algorithms to use hypotheses (i.e., models) to update and maintain a track. It is a way to do model fitting. The advantage of MHT, which help its adoption in aircraft tracking, has been the model itself. Aircraft trajectories can be easily modelled as jump linear systems. While such an assumption may be applicable to fixed-wing UAs, this would not be the case for multi-copters, which are way more flexible and versatile than traditional aircraft and helicopters, and hence are more unpredictable, by introducing non-linearities in their movement. Thus, the MHT requires extensive expert knowledge in order to create efficient and accurate models, i.e., hypotheses, to be associated to the drones.

At the same time, neural networks architectures are quite flexible and can be adapted and tailored for the specific task of early or late fusion. The issue is that these NNs (with the MLP network being well-known) are opaque in nature (i.e., they do not give us clear causal information between inputs and outputs). Furthermore, they need comprehensive (multi-modal) datasets in order to raise their trustworthiness and reliability. Alternatively, novel approaches of explainable artificial intelligence (XAI) may positively impact in this regard by providing simpler models and a way to explain the relationship between model outcomes and the input features.

It is critical to evaluate the success of DTI through a common framework of key performance indicators (KPIs), in order to compare different multi-sensor data fusion techniques. Possible KPIs would be, for each sensor type, the detection probability at different distances, e.g., 20, 50, 80, 100, 1000 meters and beyond, along with the false alarm probability (Shi et al., 2018). The lack of a uniform way to test multi-sensor, data-fusion-enabled C-UAS remains a barrier to measuring the effectiveness of such techniques.

3.3 Key challenges and the road ahead

Research on multi-sensor data fusion-enabled techniques for enhanced C-UAS implementation is limited by the following key points:

1. There is a lack of a unified framework for joint state estimation and data association identification (Zhang et al., 2024).
2. Proposals should be validated on fulfilling the architectural challenges of C-UAS, i.e., a distributed system of sensors limited by bandwidth, timing, and spatio-temporal synchronisation requirements (Wang et al., 2024a).
3. More research is needed on multi-modal range and direction fusion for accurate localisation (Strinati et al., 2024), while minimizing association with estimation errors, which leads to performance improvements (Zhang et al., 2024).
4. Specifically for the group of sensors related to electromagnetic spectrum analysis, sub 7 GHz analysis is more challenging in complex environments than mmWave, as Rician, Rayleigh, and Nakagami channel models demonstrate (Sallouha et al., 2024). However, such spectrum portion remains key for non-line-of-sight detection and tracking.
5. For machine learning, unsupervised-based multi-sensor data fusion becomes computationally intractable as the number of targets increase (Wei et al., 2024).
6. More research is needed on finding an optimal trade-off between tracking accuracy and inference time (Wang et al., 2024b). This also means building a lightweight and high-performance data processing architecture.

7. Further research should be done on reliably filter out sensor noise and bias in a heterogeneous environment ([Wang et al., 2024a](#)).
8. Currently, there is a lack of a comprehensive and high-quality dataset for multi-sensor data fusion-based C-UAS ([Wang et al., 2024a](#)).
9. Further studies are needed to achieve consensus among multiple, and potentially conflicting, sensor reports ([Li, T. et al., 2023](#)).

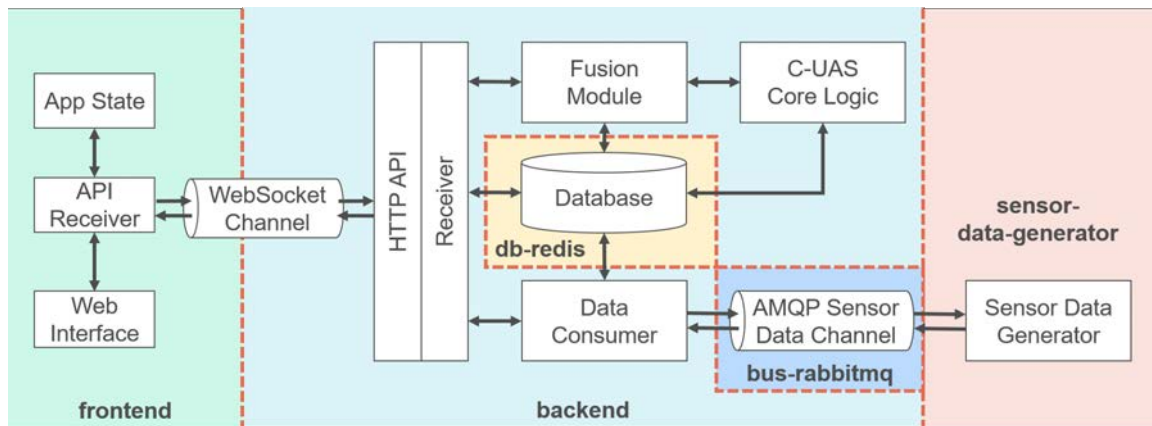
4 A Proof-of-Concept Platform for C-UAS

The full software stack that characterises the C-UAS PoC was already discussed in [European Commission \(2024a\)](#). However, the version 2.0 of the software has been overhauled to ensure the compatibility with a microservice-oriented DevOps infrastructure. While the new architecture is explained in section 4.1, the particular JRC DevOps infrastructure is described in section 4.2, in order to discuss the software continuous integration/continuous deployment (CI/CD) configuration in section 4.3. Finally, a tutorial on how to operate and maintain the new version of the software is given in section 4.4.

4.1 Architectural overview

Upon software reorganisation, the following components were isolated and decoupled from each other: (i) frontend, (ii) backend, (iii) Redis database, (iv) message bus RabbitMQ, and (v) the sensor data generator.

Figure 8. Schematic diagram of the major blocks that characterise the C-UAS PoC.



Source: JRC

The schematic diagram of Figure 8 illustrates these main components, together with a block-oriented overview of each service:

1. The frontend employs a WebSocket channel to retrieve the latest data from the backend. The application programming interface (API) Receiver is configured to be compatible with the data format of the backend, and then update the current app state. Such synchronisation mechanism serves to automatically refresh the Web Interface.
2. The backend exposes a hypertext transport protocol (HTTP)-based API with WebSocket support, in order to send the latest information that the Receiver has. The data are gathered from the fusion module, which implements the data fusion algorithm, and the data consumer, which receives the latest sensor data from the advanced message queuing protocol (AMQP)-based sensor data channel. Under the hood, the entire service logic lifecycle is handled by the C-UAS core logic, which ensures synchronisation between all these blocks that operate in a multi-threading fashion.

3. The sensor data generator produces random sensor events relatable to different UAs and sends them to the AMQP-based sensor data channel. Such service is useful for simulation and testing.

All these services are bundled into separate container images to ensure full software reproducibility, portability, and provide compatibility with the Kubernetes architecture.

4.2 The JRC's DevOps infrastructure

DevOps is a set of practices that combines software development (Dev) and IT maintenance operations (Ops) to facilitate faster and more efficient software development and deployment. Key concepts are hereby described in order to understand the C-UAS PoC integration with JRC's DevOps infrastructure:

1. Automated services – DevOps utilizes automated tools and processes to support development, testing, and release phases, reducing manual intervention and increasing efficiency.
2. Continuous Integration (CI) – In DevOps, developers frequently implement small changes, which are then quickly tested and integrated into the main codebase. This ensures that issues are caught early, and collaboration between teams is streamlined.
3. Continuous Deployment (CD) – Once a release is deemed ready, it is automatically pushed to the production environment without any manual intervention, enabling faster deployment and allowing the development team to focus on innovation.
4. Pipeline – The DevOps pipeline consists of a set of autonomous operations (configure, build, test, package, etc.) that are executed in a sequential and automated manner, ensuring consistent, reliable, and rapid releases.
5. Fast feedback on code errors and bugs – DevOps encourages quicker feedback loops, allowing developers to identify and fix issues earlier in the development process, resulting in higher-quality code and fewer production errors.
6. Allows quicker deployments and fixes from the developer – DevOps enables developers to deploy changes and fixes more rapidly, reducing the time between development and delivery, and improving the overall efficiency of the software development life cycle.

The JRC's DevOps Infrastructure is mainly composed by the following on-premises components:

1. GitLab⁴ for code sharing and CI/CD pipeline management.
2. Two Quay instances for storing publicly available⁵ and pipeline-generated⁶ container images, respectively.

⁴ <https://gitlab.jrc.ec.europa.eu> (Accessed: 16/07/2024)

⁵ <https://quay.apps.ocpt.jrc.ec.europa.eu> (Accessed: 16/07/2024)

⁶ <https://quay.apps.ocp.jrc.ec.europa.eu> (Accessed: 16/07/2024)

3. OpenShift⁷ as a Kubernetes-based container runtime environment.

GitLab is a centralised git-based repository management service, with the focus of keeping a copy of the developed software's source code in a remote server, thus facilitating sharing and synchronisation among peers. It is a well-known competitor to other similar services, such as Codeberg, BitBucket, GitHub, GitTea, and sourcehut, among others.

These services mostly rely on a role-based access control (RBAC) mechanism to access, read, write, and manage a git repository. In the case of JRC, the web authentication mechanism is handled by the Commission's active directory with multi-factor authentication (MFA) enabled. Given that such authentication mechanism is too advanced for the git protocol, a simpler, yet secure access, can be granted to git through the activation of access tokens.

A key added value of these centralised services is the complementary offering that empowers one's git repository. Indeed, special features and services can be activated according to their characteristics and requirements. For instance, a pipeline CI/CD can be defined by including Yet Another Markup Language (YAML)-formatted files in the software source code. These files will be automatically parsed by GitLab and, if compatible, will activate one or more pipeline services, which may range from code checking, spelling, dependency security scan, build, testing, and deployment, among others.

A series of ready-made templates of commonly used pipelines are available from the JRC's DevOps Wiki⁸, along with any new, updated, and other relevant information about the CI/CD infrastructure.

4.3 DevOps pipelines for unattended build and deployment

The entire pipeline relies on CI templates and basic helm charts provided by default by JRC's GitLab. This includes jobs such as code scan, dependency vulnerability check, container image build, and Kubernetes pod deployment on OpenShift. More information can be retrieved at the respective GitLab CI/CD template project⁹.

All the created Kubernetes pods are configured to not replicate¹⁰. Furthermore, each microservice is limited in CPU and runtime memory quota, to ensure proper resource usage on the Kubernetes cluster and provide reasonable limits to not cause overloads.

Regarding the backend, the git repository has a `values.yaml` and `Dockerfile` that are automatically read by the JRC's Helm chart. The goal is to configure the deployment of a podman container on OpenShift with the following specifications. The backend will connect to an AMQP-compatible broker using the address and password stored as Kubernetes secrets. The same applies to the Redis database connection. While the connection to the broker and database will be internal, the service will expose an HTTP port, which will be forwarded by the OpenShift network from a public port 80, suitable for receiving HTTP requests. An HTTP Secure (HTTPS) proxy is automatically set up by the network itself.

⁷ <https://console-openshift-console.apps.ocpt.jrc.ec.europa.eu> (Accessed: 16/07/2024)

⁸ <https://web.jrc.ec.europa.eu/devopswiki> (Accessed: 16/07/2024)

⁹ <https://gitlab.jrc.ec.europa.eu/jrc-templates> (Accessed: 17/07/2024)

¹⁰ To ensure a safe replication, a proper algorithm needs to be implemented on the microservices to cope with data races. This is left as a future work.

In a similar manner, the frontend pod is configured to be externally accessible by exposing a HTTP port, which is then served as HTTPS by OpenShift.

As for the AMQP broker, the CD pipeline is configured to use RabbitMQ “management” edition, and it is automatically configured with the Kubernetes secrets for username and password. A configuration map is created to use the stored configuration files on the git repository as runtime configuration, in order to keep all the parameters referenced and tracked by git itself. The pod will expose the AMQP port, which can be accessed only by the other pods of the project.

Similar to the AMQP broker, the Redis database pod is configured by following the same logic, i.e., (i) use the “Redis stack server” container image, (ii) which it is configured by using Kubernetes secrets, and (iii) the database connection port is exposed for internal services.

Finally, the sensor data generator pod is configured to communicate with the AMQP broker.

4.4 Tutorial: how to develop and deploy an update of C-UAS PoC

The entire software heavily relies on the use of git as a tool to manage and synchronise the source code to both GitLab and peers. To this end, it is essential to learn more on how git works¹¹. This program needs further configuration to ensure that the tool adheres to the security standards imposed by the JRC’s DevOps infrastructure. Furthermore, instructions on how to clone the project, modify, check the edits, and then commit and push are given. In order to ensure proper reproducibility, it is recommended to download a recent Linux-based distribution, such as Fedora¹², with at least the following programs installed:

- git version 2.44 or greater
- podman version 4 or greater
- podman-compose version 1.2 or greater
- podman-docker version 4 or greater
- make version 4 or greater.

4.4.1 How to configure git authentication for JRC’s GitLab

Git can be configured to use GitLab’s generated Personal Access Tokens. These are scope and time-limited tokens that can be used to allow HTTPS-based authentication between a git command executed on the workstation and GitLab. A new token can be created and git can be instructed to use it by following GitLab’s official guide¹³.

Apart from authentication, commit signing is also mandatory in JRC’s GitLab, i.e., all git commits done by a developer must be signed with a GPG key that is trusted on GitLab and kept secret by the

¹¹ https://docs.gitlab.com/ee/tutorials/learn_git.html (Accessed: 16/07/2024)

¹² <https://fedoraproject.org> (Accessed: 16/07/2024)

¹³ https://docs.gitlab.com/ee/user/profile/personal_access_tokens.html (Accessed: 16/07/2024)

developer. To this end, the official GitLab guide¹⁴ offers some instructions on how to create a GPG private/public key pair and upload the public one on the website.

4.4.2 Clone the project main repository

“CUAS-DevEnv” represents the main entry of the developer, to quickly get the full environment and all the microservices, ready for improvement and testing. To ensure proper project cloning, it is essential to use the flag `--recurse` of `git`, to ensure that the `git` submodules are cloned. “`git-clone`” manual¹⁵ and the `git` book chapter on submodules¹⁶ can clarify what and how to do it.

4.4.3 Run the project locally

The entire software stack can be prepared and run through the `make` command. To start, `make pull` and `make build` allow to download all the required base container images and build the latest version of project’s container images, respectively. During the build phase, each microservice code is compiled and packaged in a container image. Finally, `make up` starts all the containers. It is possible to check the log messages of each service through `podman compose logs` command. More information on how to effectively use this command is available on the official manual¹⁷.

4.4.4 Edit the project, commit changes, and activate CI/CD pipelines

It is possible to edit project files via any text program, code editor, or integrated development environment. Once changes are made, `make build` can be executed to update the container images and see the effects locally. Finally, once the changes are satisfying, these can be staged on `git` to create a commit. More information on how staging and commit work can be found on the `git` official book¹⁸.

4.4.5 Clean up the environment

Before deleting the project directory, it is recommended to execute the command `make rmi`, which will clear all the built container images.

¹⁴ https://docs.gitlab.com/ee/user/project/repository/signed_commits/gpg.html (Accessed: 16/07/2024)

¹⁵ <https://git-scm.com/docs/git-clone> (Accessed: 16/07/2024)

¹⁶ <https://git-scm.com/book/en/v2/Git-Tools-Submodules> (Accessed: 16/07/2024)

¹⁷ <https://docs.podman.io/en/latest/markdown/podman-compose.1.html> (Accessed: 16/07/2024)

¹⁸ <https://git-scm.com/book/en/v2/Git-Basics-Recording-Changes-to-the-Repository> (Accessed: 16/07/2024)

5 Conclusions

With the emerging use cases of the society that mandate personalised, faster, safer, and more reliable and secure cyber-physical systems, UAS are becoming key enablers for the re-evaluation of transport, communication, and monitoring solutions. Together with the new ecosystem of technologies proposed by 6G networks, UAs have the potential to be employed for large-scale BVLoS missions, traversing rural and urban areas, and thus simple and complex environments, which poses challenges for both the physical and the cyber worlds. As with all technologies, their improvement brings tremendous benefit for the society, but also new types of threats, which should be readily discussed and mitigated through proper defensive system, to protect society and institutions from the misuse of drones by malicious actors.

To this end, this report provided an overview of current C-UAS solutions in EU-funded endeavours and scientific literature. Furthermore, a taxonomy of sensors used in these systems is provided, which is crucial to understand the open challenges for developing reliable, robust, and accurate C-UAS solutions, and how more research into multi-sensor data fusion may address them. a meta-study analysis of the scientific literature is presented concerning the state of the art and challenges regarding multi-sensor data fusion algorithms for C-UAS. The study has been limited to the DTI task, considering also the classification of UAS, as it fits the scope of C-UAS employed for civil purposes. As part of JRC's efforts in this regard, it is presented an update to the C-UAS PoC, including an architectural overview, organizational structure, and an overview of the JRC's DevOps infrastructure. The DevOps pipelines for unattended build and deployment are described, and a tutorial is provided on how to develop and maintain the entire software.

To continue this work, some key ideas are hereby suggested as future works:

1. Design and implement an algorithm-agnostic benchmark framework for multi-target tracking and a common set of key performance indicators.
2. Pursue a state of the art and an in-depth comparison of multi-UAV tracking via multi-sensor C-UAS.
3. Propose of a comprehensive C-UAS solution that offers a guideline to both research and industry for future endeavours.

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List of abbreviations and definitions

AMQP	advanced message queuing protocol
API	application programming interface
ATR	automatic target recognition
BVLoS	beyond visual line of sight
CI/CD	continuous integration/continuous deployment
CSI	channel state information
C-UAS	counter-unmanned aircraft system
DL	deep learning
DoA	direction of arrival
DTI	detection, tracking, and identification
FAA	Federal Aviation Administration
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HAP	high-altitude platform
HTTP	hypertext transport protocol
HTTPS	HTTP secure
IMM	interacting multiple model
IoT	internet of things
IR	infra-red
JRC	Joint Research Centre
KPI	key performance indicator
LASER	light amplification by stimulated emission of radiation
LiDAR	light detection and ranging
MFA	multi-factor authentication
MHT	multi-hypothesis tracking
NLR	Netherlands Aerospace Centre
NTN	non-terrestrial network
PDA	probabilistic data association
PoC	proof of concept

PTZ	pan-tilt-zoom
RADAR	radio detection and ranging
RBAC	rely on a Role-based Access Control
RF	radiofrequency
RPAS	remotely piloted aircraft systems
SAR	synthetic aperture RADAR
SDR	software defined radio
SONAR	sound navigation and ranging
SWIR	short-wave IR
TDoA	time delay of arrival
TN	terrestrial network
UA	unmanned aircraft
UAS	unmanned aircraft system
UAV	unmanned aerial vehicle
UTM	unmanned traffic management
YAML	yet another markup language
XAI	explainable AI

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